

[PYTHON ESSENTIALS]

NAME :- ABHIJEET PANDEY

**COURSE :- BTECH (ELECTRICAL
ENGINEERING)**

**SUBJECT :- PROGRAMMING FUNDAMENTALS
FOR ELECTRONICS ENGINEERS**

ADMISSION NUMBER :- 25SECE1050025

1.Accept Name and Age and Display Data Types

#CODE

```
name = input("Enter your name: ")
age = int(input("Enter your age: "))

print(f"Hello {name}, you are {age} years old.")

# Display data types
print("Data type of name:", type(name))
print("Data type of age:", type(age))
```

#OUTPUT

```
Enter your name: Abhijeet
Enter your age: 18
Hello Abhijeet, you are 18 years old.
Data type of name: <class 'str'>
Data type of age: <class 'int'>
```

2.Program to Check Positive, Negative, or Zero

#code

```
num = float(input("Enter a number: "))
```

```
if num > 0:
```

```
    print("The number is Positive.")
```

```
elif num < 0:
```

```
    print("The number is Negative.")
```

```
else:
```

```
    print("The number is Zero.")
```

#OUTPUT

```
Enter a number: 6
The number is Positive.
```

```
===== RESTART: C:/Users/pande/Downloads/prctice.py =====
```

```
Enter a number: -1
The number is Negative.
```

```
===== RESTART: C:/Users/pande/Downloads/prctice.py =====
```

```
Enter a number: 0
The number is Zero.
```

```
|
```

3. Program to Calculate Sum of First N Natural Numbers

#code

```
n = int(input("Enter a number: "))
```

```
total = 0
```

```
for i in range(1, n + 1):  
    total += i
```

```
print("Sum of first", n, "natural numbers is:", total)
```

#OUTPUT

```
===== RESTART: C:/Users/pande/Downloads/prctice.py =====  
Enter a number: 5  
Sum of first 5 natural numbers is: 15
```

4. Program Demonstrating Arithmetic, Comparison, and Logical Operators

#code

```
a = int(input("Enter first number: "))
b = int(input("Enter second number: "))

print("Addition:", a + b)
print("Subtraction:", a - b)
print("Multiplication:", a * b)
print("Division:", a / b)

print("a > b:", a > b)
print("a < b:", a < b)
print("a == b:", a == b)

print("(a > 0 and b > 0):", (a > 0 and b > 0))
print("(a > 0 or b > 0):", (a > 0 or b > 0))
print("not(a > 0):", not(a > 0))
```

#OUTPUT

```
Enter first number: 12
Enter second number: 6
Addition: 18
Subtraction: 6
Multiplication: 72
Division: 2.0
a > b: True
a < b: False
a == b: False
(a > 0 and b > 0): True
(a > 0 or b > 0): True
not(a > 0): False
```

5. Program to Find Maximum, Minimum, Sum, and Sort a List

#code

```
numbers = list(map(int, input("Enter numbers separated by
space: ").split()))
```

```
print("Maximum number:", max(numbers))
```

```
print("Minimum number:", min(numbers))
```

```
print("Sum of numbers:", sum(numbers))
```

```
numbers.sort()
```

```
print("Sorted list:", numbers)
```

#OUTPUT

```
----- RESTAINT: C:/Users/pamuc/L
Enter numbers separated by space: 1 2 3 4 5
Maximum number: 5
Minimum number: 1
Sum of numbers: 15
Sorted list: [1, 2, 3, 4, 5]
```

6. Program for String Operations

```
#code
```

```
text = input("Enter a string: ")
```

```
vowels = "aeiouAEIOU"
```

```
count = 0
```

```
for char in text:
```

```
    if char in vowels:
```

```
        count += 1
```

```
reverse_text = text[::-1]
```

```
if text == reverse_text:
```

```
    palindrome = "Yes"
```

```
else:
```

```
    palindrome = "No"
```

```
print("Number of vowels:", count)
print("Reversed string:", reverse_text)
print("Is palindrome?", palindrome)
```

#OUTPUT

```
Enter a string: madam
Number of vowels: 2
Reversed string: madam
Is palindrome? Yes

===== RESTART: C:/Users/pande/Downloads/prctice.py =====
Enter a string: abhijeet
Number of vowels: 4
Reversed string: tejihba
Is palindrome? No
```

7. Program Using Functions for Factorial and Fibonacci Sequence

#code

```
def factorial(n):
    fact = 1
    for i in range(1, n + 1):
        fact *= i
    return fact

def fibonacci(n):
    a, b = 0, 1
    print("Fibonacci Sequence:")

    for i in range(n):
        print(a, end=" ")
        a, b = b, a + b
```



```
num = int(input("Enter a number: "))
```

```
print("Factorial =", factorial(num))
```

```
fibonacci(num)
```

#OUTPUT

```
Enter a number: 6
Factorial = 720
Fibonacci Sequence:
0 1 1 2 3 5
|
```

8. Program Using Dictionary for Student Marks

#code

```
students = {}
```

```
n = int(input("Enter number of students: "))
```

```
for i in range(n):
```

```
    name = input("Enter student name: ")
```

```
    marks = int(input("Enter marks: "))
```

```
    students[name] = marks
```

```
highest = max(students, key=students.get)
```

```
lowest = min(students, key=students.get)
```

```
print("Student Records:", students)
```

```
print("Highest Scorer:", highest, "-", students[highest])
```

```
print("Lowest Scorer:", lowest, "-", students[lowest])
```

#OUTPUT

```
Enter number of students: 5
Enter student name: Abhijeet
Enter marks: 85
Enter student name: Arman
Enter marks: 87
Enter student name: Ajay
Enter marks: 90
Enter student name: Bhuvan
Enter marks: 82
Enter student name: Daksh
Enter marks: 80
Student Records: {'Abhijeet': 85, 'Arman': 87, 'Ajay': 90, 'Bhuvan': 82, 'Daksh': 80}
Highest Scorer: Ajay - 90
Lowest Scorer: Daksh - 80
```

9. Program to Read File, Count Words and Lines with Exception Handling

#code

try:

 filename = input("Enter file name: ")

 with open(filename, 'r') as file:

 content = file.read()

 lines = content.split('\n')

 line_count = len(lines)

 words = content.split()

 word_count = len(words)

 print("Number of lines:", line_count)

 print("Number of words:", word_count)

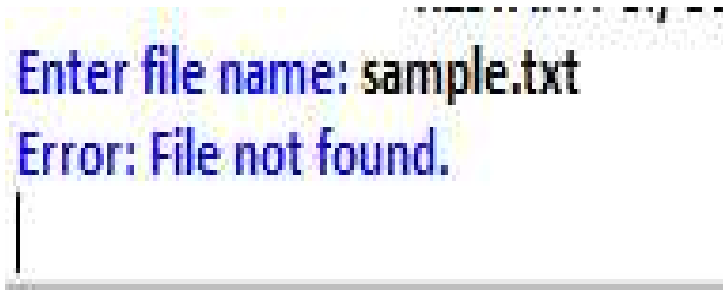
except FileNotFoundError:

```
print("Error: File not found.")
```

```
except Exception as e:
```

```
    print("An error occurred:", e)
```

#OUTPUT

A terminal window screenshot with a dark background. The first line shows the prompt 'Enter file name: ' followed by the user input 'sample.txt'. The second line shows the output 'Error: File not found.' followed by a cursor. A horizontal line is visible at the bottom of the terminal window.

```
Enter file name: sample.txt
Error: File not found.
```